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BCAAIMS 102

I Semester B.C.A. (AI and ML) Degree Examination, December 2024/January 2025
(SEP) (2024 – 25 Batch Onwards)
PROBLEM SOLVING USING C

Time : 3 Hours Max. Marks : 80

Note : *Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.*

PART – A

1. a) Define algorithm. **(10×2=20)**
   b) List any two important features of C.
   c) List any four keywords.
   d) What is the difference between = and == operators ?
   e) Give the syntax of conditional operator. Give an example.
   f) List any four built-in Mathematical functions available in C.
   g) Differentiate entry-controlled and exit-controlled loops.
   h) Write a statement that declares an integer array of 10 element size and initializes it with values.
   i) List any four character handling functions.
   j) What is pointer ? Specify its use.
   k) What is a recursive function ?
   l) How a file can be closed ?

PART – B
Unit – I

2. a) Explain structure of C program with example.
   b) Write a note on Constants in 'C'.
   c) Explain with examples declaring and assigning value to variables. **(6+5+4)**

P.T.O.

3. a) Discuss the benefits and limitations of flowcharts.
   b) List and explain primary data types available in 'C'.
   c) Explain with example #define statement in 'C'. (6+5+4)

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Unit – II

4. a) Explain different forms of if statement.
   b) Explain with examples formatted input in C.
   c) Explain the concept of type conversion in C. Give examples. (6+5+4)

5. a) List and explain arithmetic, logical and relational operators available in C.
   b) Explain switch statement with syntax and example.
   c) Explain printf() function with its syntax and example. (6+5+4)

Unit – III

6. a) Explain with example how to declare, initialize and use strings in C.
   b) Explain with example any two categories of user defined functions.
   c) Write a program to sort elements of an integer array. (6+5+4)

7. a) Explain one-dimensional array with the help of suitable example.
   b) List and explain any five-string handling functions available in C.
   c) Explain for statement with syntax and example. (6+5+4)

Unit – IV

8. a) Explain with example defining a structure, declaring structure variable and accessing structure members.
   b) Write a note on pointer increments and scale factor.
   c) Explain the following file handling functions in 'C' with syntax.
      i) getc()
      ii) putc()
      iii) getw()
      iv) putw() (6+5+4)

9. a) Explain with example defining a union, declaring union variable and accessing union members.
   b) What is a file ? Explain any four types of file access modes.
   c) Explain how pointers are used to access string ? (6+5+4)